

CALVIN (HWALANG) CHO

Email: calvincho23@gmail.com | Phone: 714-308-6247 | [LinkedIn](#) | [Personal Website](#)

EDUCATION

Duke University 2022 - 2026
Bachelor of Arts in **Biology and Computer Science** 3.5 / 4.0

- **Activities:** Student Collaborative on Health Policy, iGEM (International Genetically Engineered Machine), Undergraduate Research Support Office Student Advisory Board, International Students Center Global Fellowship
- **Relevant Coursework:** Probability, Machine Learning, Data Structures & Algorithms, Database Systems, Computer Systems, Discrete Math

RESEARCH EXPERIENCE

Columbia University Irving Medical Center, Department of Systems Biology May 2026 - present
Summer Research Intern
Advisor(s): Raul Rabadan, PhD; Anqi Wang, PhD

- Developing computational pipelines and statistical methods for investigating HLA-I alternative splicing using TCGA RNA-seq data with controlled-access de-identified datasets

National Aeronautics and Space Administration (NASA) August 2025 - present
AI/ML Analysis Working Group Member, GeneLab

- Developing machine learning models to analyze >1M single-cell perturb-seq profiles and integrate multi-omics data for astronaut health applications
- Designing predictive frameworks to classify thousands of genetic perturbations and inform countermeasure design for long-duration space missions

Duke University School of Medicine, Department of Neurology September 2022 - May 2026
Undergraduate Research Assistant
Advisor(s): Carlene Moore, PhD; Malak Fouani, PhD

- Developed and designed automated imaging pipeline integrating Cellpose segmentation, Z-projections, and channel alignment, reducing analysis time from *8 minutes to 6 seconds per image* (>99% efficiency gain)
- Analyzed 5,000+ images from immunocytochemistry and calcium imaging experiments to investigate the role of TRPV4 ion channels and P2Y receptors in migraines/headaches, ongoing projects involving assays, clinical trials (Botox), cell culture, and mice dissection

University of California, San Francisco Innovation Ventures June 2025 - August 2025
Summer Catalyst Intern
Advisor(s): Julian Motzkin, MD, PhD; Prasad Shirvalkar, MD, PhD

- Designed a 45-field diagnostic evaluation framework to benchmark clinical performance, regulatory readiness, and commercialization potential
- Conducted comparative analysis of 20+ competitor solutions across 9 regulatory pathways, identifying gaps in clinical utility and market positioning

United States Department of Justice, Bureau of Justice Statistics August 2024 - October 2024
Student Expo Program Researcher / Presenter
Advisor(s): Matt Hickman, PhD; Matt Durose, MS

- Analyzed the Census of Publicly Funded Forensic Crime Laboratories, 2020 dataset (538 variables across federal, state, county, and municipal labs) using R
- Identified disparities in competency results, with federal labs averaging 93.9% compared to lower ratings in state, county, and municipal labs

Regeneron Pharmaceutical, Cardiovascular Diseases Therapeutics Focus May 2024 - August 2024
Summer Research Intern
Advisor(s): Feng Luo

- Conducted 8 experiments (ELISA, MSD, Gyroslab) to assess edema side effects of tissue plasminogen

- activator (tPA) treatment for ischemic stroke and tested antibodies (anti-FXII) for therapy use
- Created 50-slide presentation on potential utilization of Regeneron antibodies as treatment for tPA side effects

TECHNICAL SKILLS

- **Computational:** Python, R, machine learning (elastic net, random forest, gradient boosting, Bayesian inference), RNA-seq analysis, single-cell genomics, multi-omics data integration, time-series analysis, statistical modeling, image analysis (Cellpose, EBIImage), pipeline development
- **Laboratory:** ELISA, MSD, Gyroslab, immunocytochemistry, calcium imaging, cell culture, tissue dissection, GLP compliance
- **Tools & Platforms:** Git, HPC/cluster computing, TCGA controlled-access datasets, NASA GeneLab, R Shiny, Microsoft Office

PUBLICATIONS

Journal Articles

Fouani M, Kumari S, Charles A, Wickware C, Moore A, Cho C, Moore C. "The TRPV4-Mast Cell Axis: Implications for Neurogenic Inflammation and Chronic Disease." *International Journal of Molecular Sciences*. March 2026.

Conference Workshop Papers

Cho C. "Biological Hallucinations in Time-Series Foundation Models: A Benchmark on High-Frequency Neural Waveforms." *ICML 2026 Workshop on Foundation Models for Structured Data (FMSD)*. Seoul, South Korea, 2026.

Cho C. "Proteomic Divergence in the Trisomic Mouse Cortex: Machine Learning Identifies Tau, APP, and ADARB1 as Key Genotype Signatures and Reveals Limited Proteomic Response to Memantine." *ICML 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GenBio)*. Seoul, South Korea, 2026.

Submitted Manuscripts

Werner K, Wickware C, Moore AA, Cho C, Moore D, Collins T, Moore C. "Biomarker Study in Children and Adolescents with New Daily Persistent Headache (NDPH)." Submitted to *Cephalalgia*, December 2025.

Manuscripts in Preparation

Sengupta S, Rozen S, Cho C, et al. "Anti-CGRP Neutralizing Antibody for Modulation of Neurogenic Inflammation in Trigeminal and Glossopharyngeal Pain Associated with Small Fiber Neuropathy/Fibromyalgia, a Phase 1B Clinical Study." In preparation, 2026.

SCIENTIFIC PRESENTATIONS

Cho C. Poster Presentations. *International Conference on Machine Learning (ICML) GenBio and FMSD workshops*, Seoul, South Korea, 2026.

Cho C, et al. Poster Presentation. *Duke Department of Neurology TBS/DCEC Research Symposium*. Durham, NC, 2025.

Cho C, et al. Poster Presentation. *R Foundation useR! Conference*. Durham, NC, 2025.

Cho C, et al. Poster Presentation. *Duke Annual Undergraduate Research Symposium*. Durham, NC, 2024.

HONORS / AWARDS

Reagan Institute Civic Leaders Fellowship, Ronald Reagan Presidential Foundation & Institute	2025
Journal Scholars Exchange, American Statistical Association (ASA) & AAPOR	2024
Health Policy Case Competition Finalist (top 3 / 40), Duke University	2023 & 2024
Health Policy Case Competition Finalist (top 3 / 88), Tulane University	2023